Pathology: Nephrotic Syndrome

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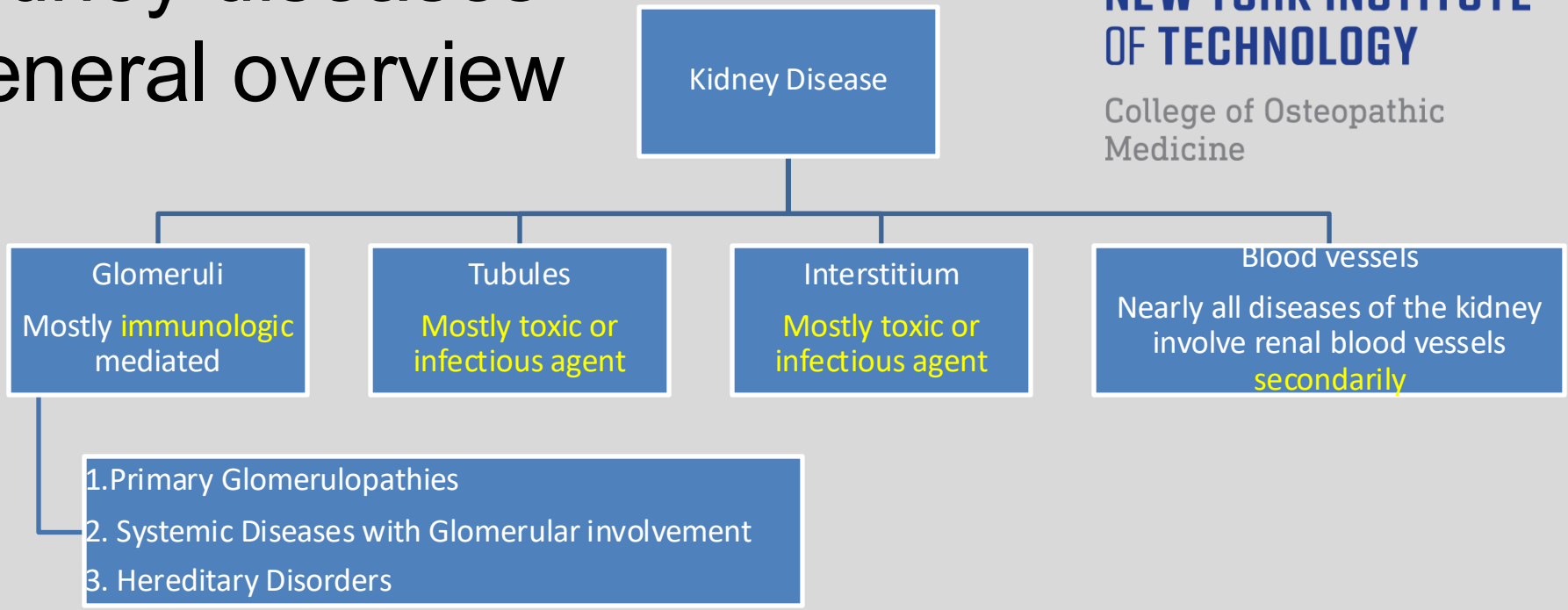
Session Objectives

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1. Following a brief review of renal pathophysiology, interpret the pathogenesis, pathophysiology, pathology and clinicopathology of minimal change disease (**MCD**).
2. Identify the pathogenesis, pathophysiology, pathology and clinicopathology of focal segmental glomerulosclerosis (**FSGS**)
3. Analyze the pathogenesis, pathophysiology, pathology and clinicopathology of membranous glomerulonephritis (**MGN**).
4. Contrast the pathogenesis, pathophysiology, pathology and clinicopathology of **diabetic nephropathy** and **renal amyloidosis**.
5. Identify the pathogenesis, pathophysiology, pathology and clinicopathology of membranoproliferative glomerulonephritis (**MPGN**).

Kidney diseases general overview

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Nephrotic Syndrome

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- ↑ permeability of **GBM** to protein
  - The nephrotic syndrome has diverse causes
    - Share a common pathophysiology
    - Derangement in the capillary walls of the glomeruli that results in **↑ permeability to plasma proteins**

→ Proteinuria, **3.5g daily** (**Foamy** Urine)

→ ↓Antithrombin III → Hypercoagulable state

→ ↓Immunoglobulins → Infection

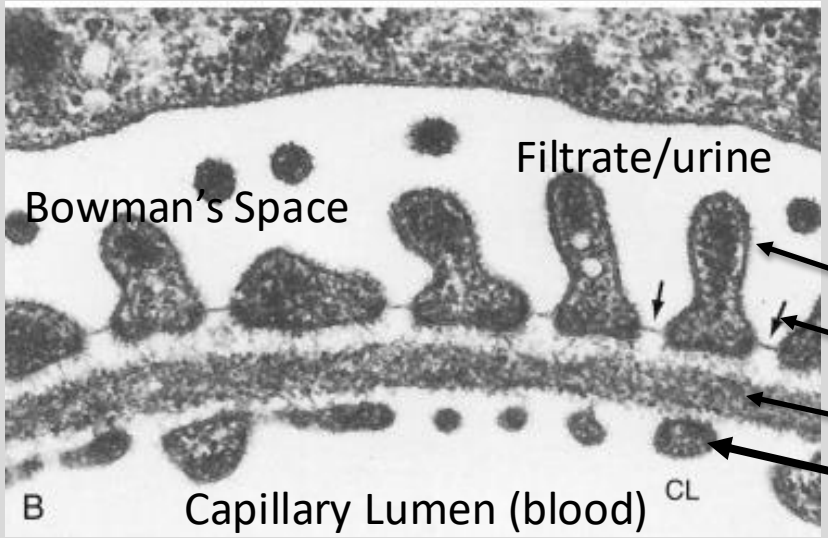
→ ↓Albumin → ↑ Synthesis of lipoproteins in liver → Hyperlipidemia

→ ↓Albumin → ↓Plasma Oncotic Pressure → Edema

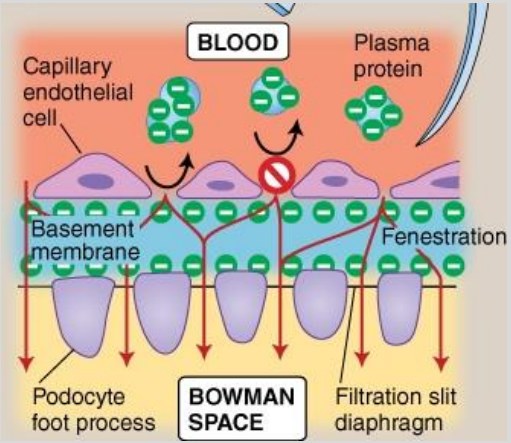
- In the face of continuing proteinuria, salt & water retention further aggravates **edema**, if unchecked may lead to development of generalized edema (termed **anasarca**)
- At the **onset**, there is little or **no azotemia** or **hypertension**

Glomerular Filtration Barrier-EM

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Podocyte foot process
Filtration slit
Basement Membrane
Fenestrated Capillary Endothelium



- Recall **negative charge barrier to proteins**: In patients w/ nephrotic syndrome, negative charge barrier of glomerular basement membrane & podocytes is **disrupted**

- Gartner, Leslie P. Color Atlas and Text of Histology 7e ch16
- Preston, Robin R., Wilson, Thad E. Lippincott's Illustrated Reviews: Physiology 1e Ch. 25

Minimal Change Disease (MCD)

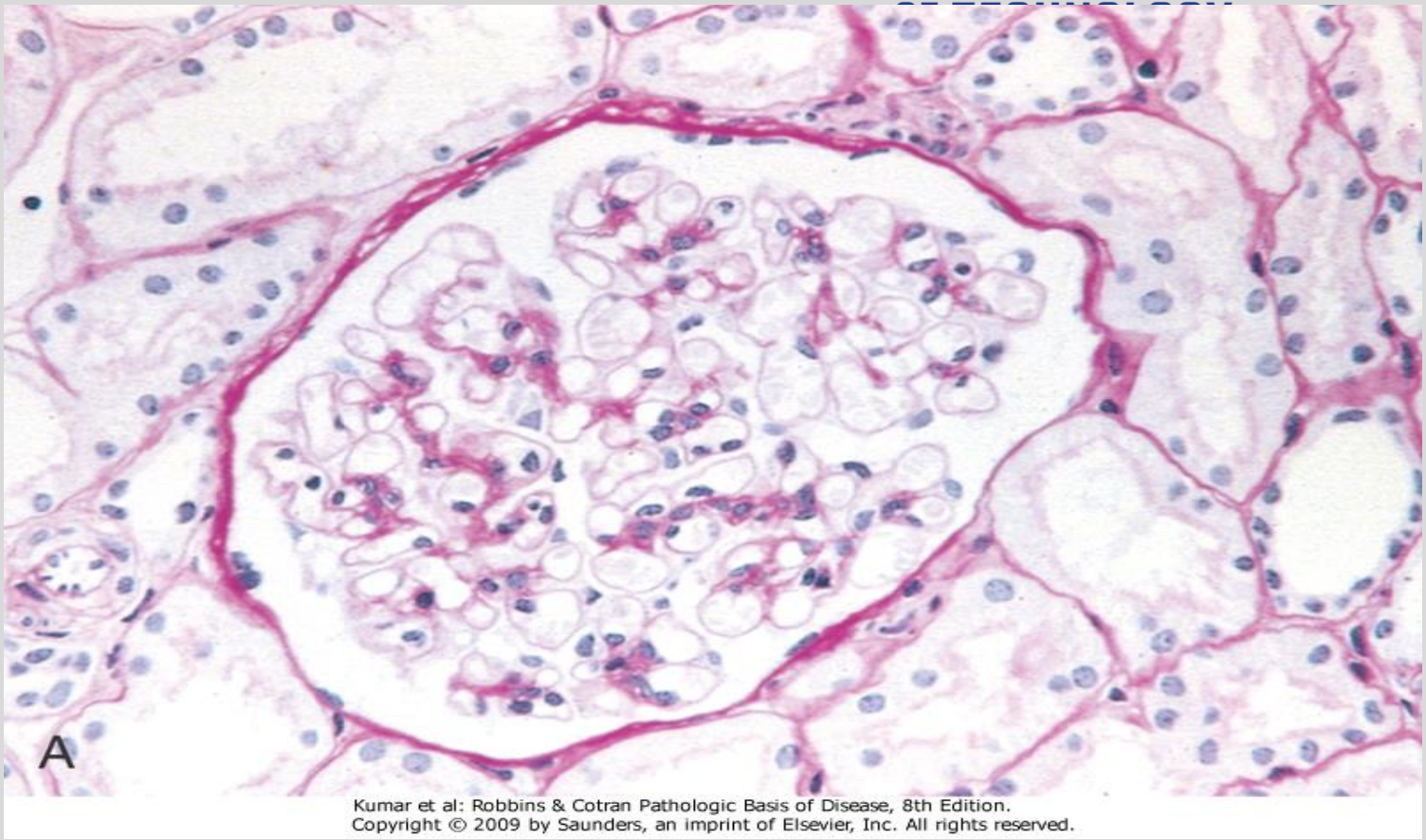
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- a.k.a. Lipoid Nephrosis, Nil Disease
- Most common cause of nephrotic syndrome in **children**
- Associated with a **recent infection** or **immunization**
  - Other causes:
    - **NSAIDS**
    - Lymphomas (increased incidence in **Hodgkin** disease)
    - Leukemias
- Pathogenesis: **T-cell cytokines** trigger damage to foot processes of **podocytes** → Selective proteinuria (**albumin** is the primary protein lost)

- Light microscopy (**LM**): Glomeruli appear **normal**, positive **fat stains** in glomerulus & tubules
- Immunofluorescence (IF) microscopy: Negative
- Electron microscopy (**EM**): Podocytes – loss of foot processes w/ swelling, fusion of those remaining-**foot process effacement** & **no deposits**

MCD LM

- PAS stain shows **basement membranes** well
- Glomerulus looks **normal**

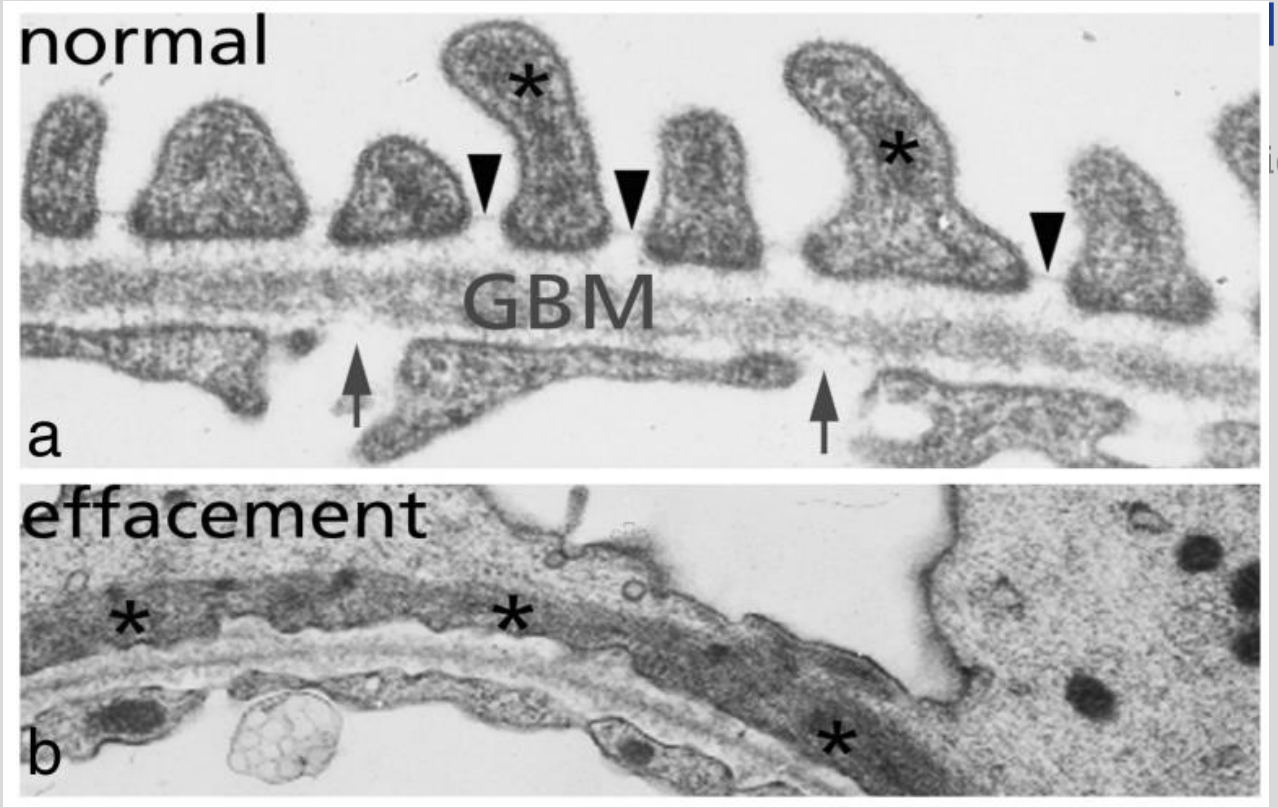


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MCD EM

- Podocyte foot processes* are **fused** & **effaced**

- **GBM** *

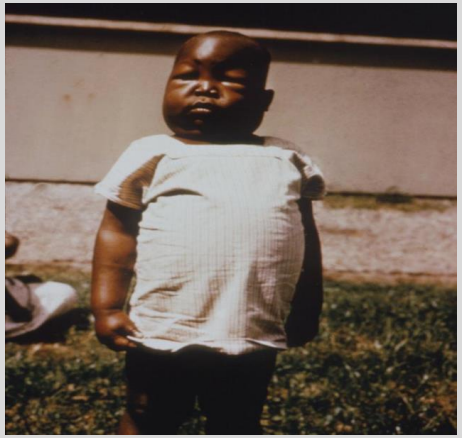


MCD Clinical Presentation

- **7-year-old** boy is brought in by mother because of **diffuse swelling** of body and "**foamy urine**".
  - ROS: (+) **Recent URI**
  - UA: (+) **Massive protein**
  - PE: (+) **Anasarca**, (+) CVA tenderness of R flank, **normotensive** (**compared** to other types of nephrotic syndromes)
  - **Responds** well to **steroid** therapy

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Anasarca

Focal Segmental Glomerulosclerosis (FSGS)

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- A disease and pattern: **Focal =** only **some glomeruli** involved
  - **Segmental =** only **part** of the **glomerulus**
- **Hispanic & African Americans** > Caucasians
- Most common cause of nephrotic syndrome in **adults** in US

- Types & associations:
  - Primary (idiopathic)
  - Secondary: **4 H's**
    - HIV (**Collapsing** glomerulopathy)
    - Heroin use (IV)
    - Hemoglobinopathy (Sickle cell disease)
    - Heavy (Massive **obesity**)

- Pathogenesis: Degeneration & focal disruption of **podocytes** + **sclerosis** w/ **hyalinosis**

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FSGS

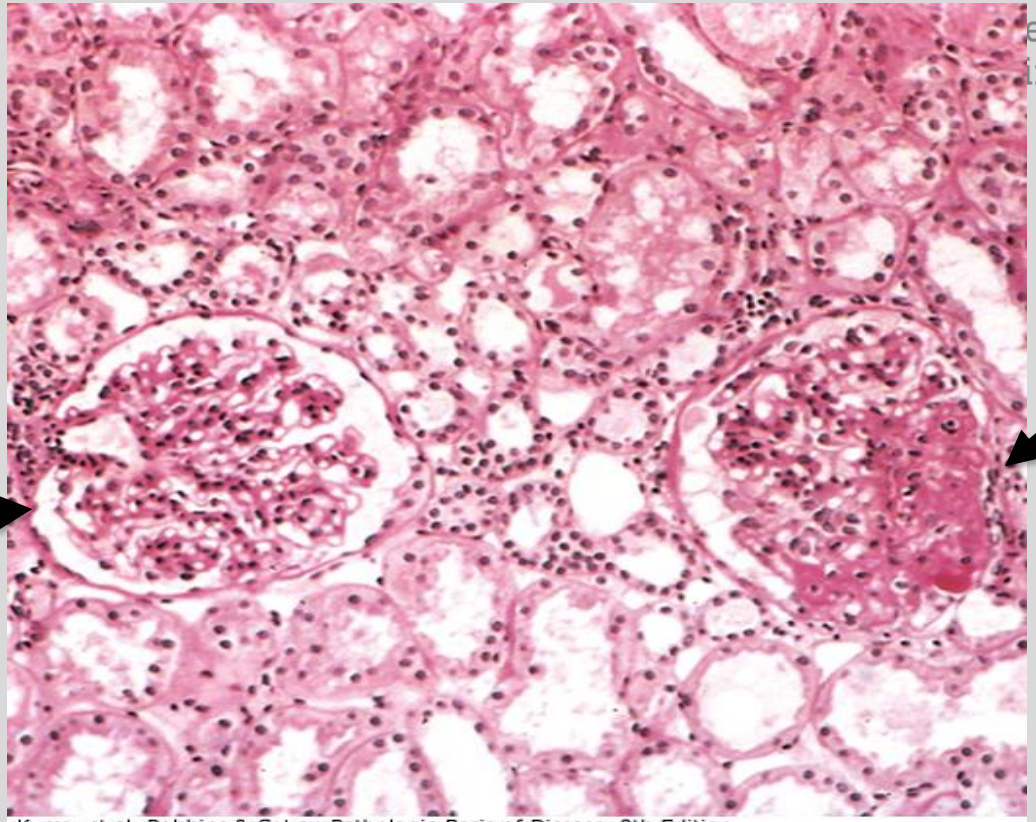
- Pathology
  - LM: FSGS w/ **hyalinosis**
  - IF: Usually **negative**; maybe non-specific IgM, C3 entrapment in areas of scarring
  - EM: Fusion, **effacement** of podocytes, **no** electron dense **deposits**

FSGS LM

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H&E tissue stain, nuclei blueish, cytoplasm pink-red

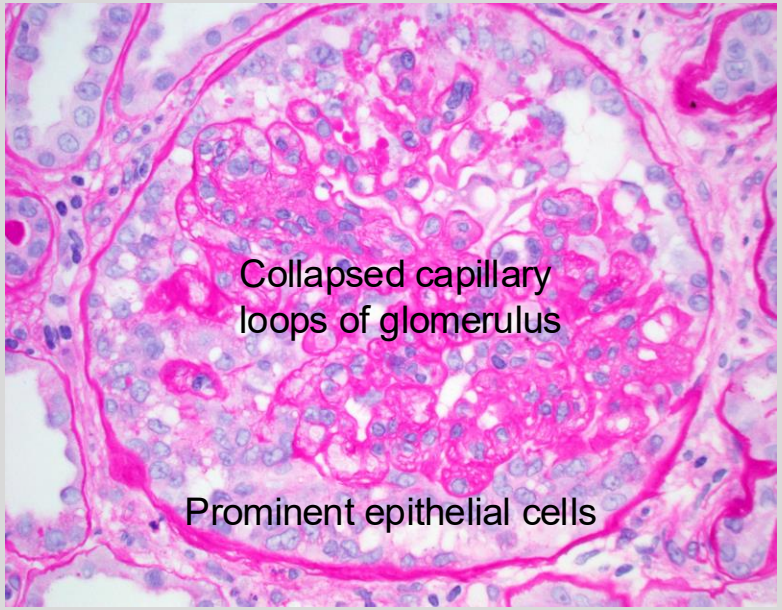


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FSGS collapsing variant

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- PAS stain
  - Collapse of the glomerular tuft & prominence of the overlying podocytes or parietal epithelial cells
  - Idiopathic or associated w/ HIV, drug toxicities & microvascular injuries
  - **Worse** prognosis than other variants of FSGS

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FSGS clinical presentation

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- **Differences** from minimal change disease:
  - Higher incidence of: **Hematuria** (microscopic), **hypertension**
  - **NON-selective** proteinuria
  - **Poor response** to corticosteroid therapy
  - **50%** of patients develop **ESRD within 10 years**

- Example presentation
  - **44-year-old** homeless, IV drug (**heroin**) user presents with **foamy** urine & diffuse **edema**
    - Lab: ↑ serum **creatinine** at presentation
    - UA: Nephrotic range **proteinuria**, (+) small **blood**, (+) small **leukocytes**

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Membranous Glomerulonephritis (MGN)/Nephropathy

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- M:F **2:1**
- Usual age presentation range: **30-50 years old**
- Types:
  - **Primary**/idiopathic
    - Caused most likely by antibodies to a renal autoantigen ~ 70% of primary is due to **phospholipase A_2 receptor antibodies**
  - Secondary
    - Drugs: captopril, **NSAIDS**, penicillamine
    - Infections: **HBV**, **HCV**, Plasmodium malariae, syphilis
    - Malignancy: **carcinomas (solid tumors)** of lung & colon, melanoma
    - Autoimmune disease: **SLE** (nephrotic presentation- **class V**), thyroiditis

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MGN/Nephropathy pathogenesis & pathology

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- **Chronic** immune complex-mediated disease
- In-situ antigen:antibody (Ab:Ag interaction; **Ab:PLA2R**)
- Mechanism for leaky glomerular capillary wall:
  - **Subepithelial** immune deposits activate complement→ **C5b-C9 (MAC)**→Podocyte injury→ **NON-selective** proteinuria

- LM: Diffuse thickening of **capillary & GBM** due to subepithelial Ig deposits
- IF: **Granular** pattern (in capillary walls - **IgG,C3**)
- EM: **Effacement**/Fusion of foot processes, subepithelial electron dense deposits (**spike & dome pattern)**

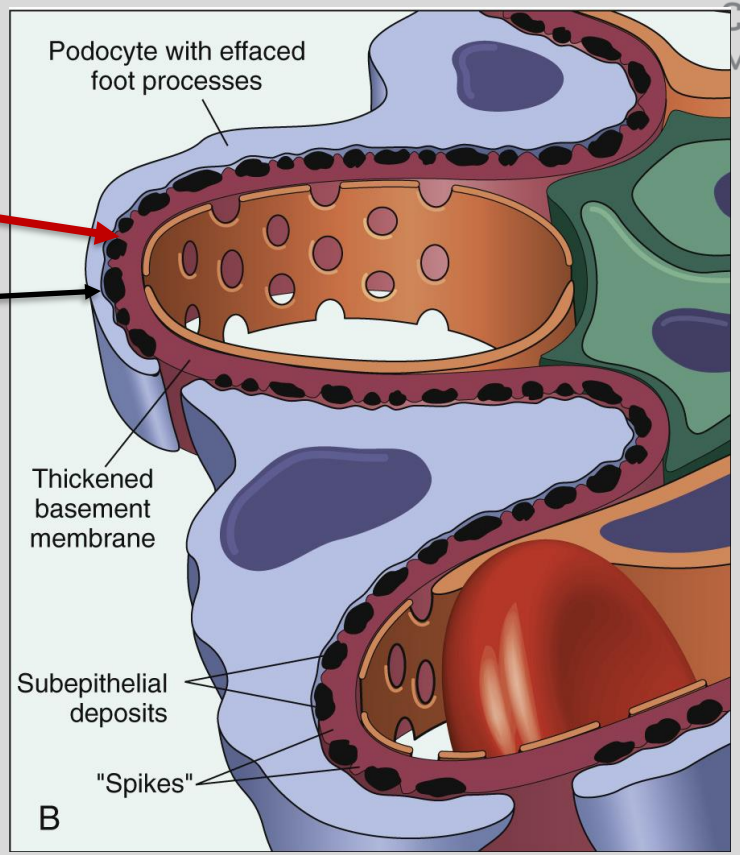
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MGN/Nephropathy

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Spike: GBM reaction to immune complex deposit

Dome: Immune complex thicken GBM

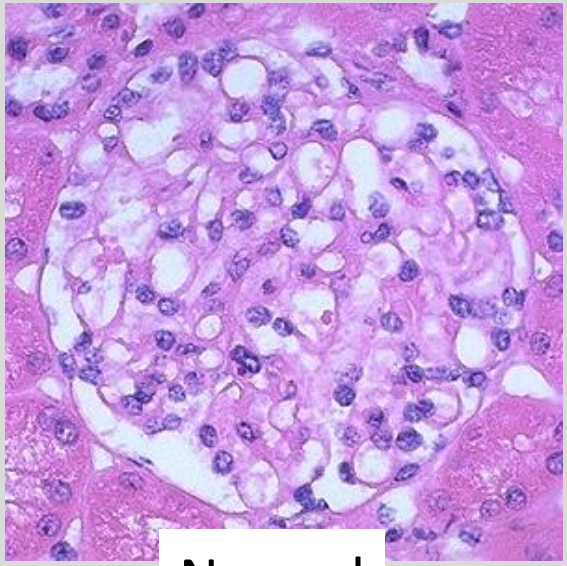


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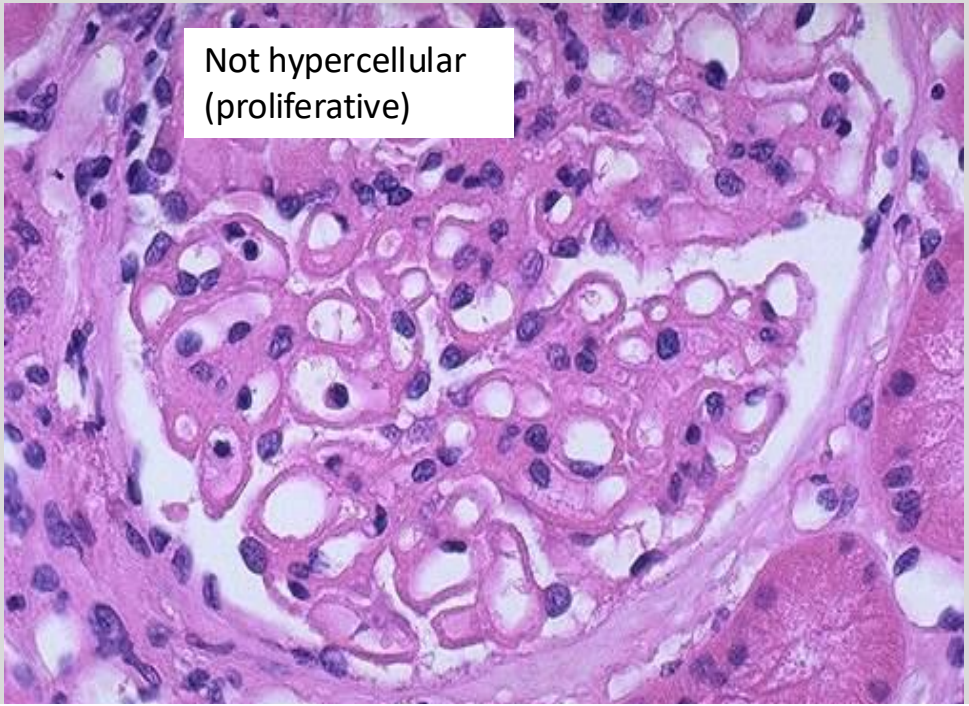
MGN/Nephropathy LM

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H&Es: Glomeruli



Normal

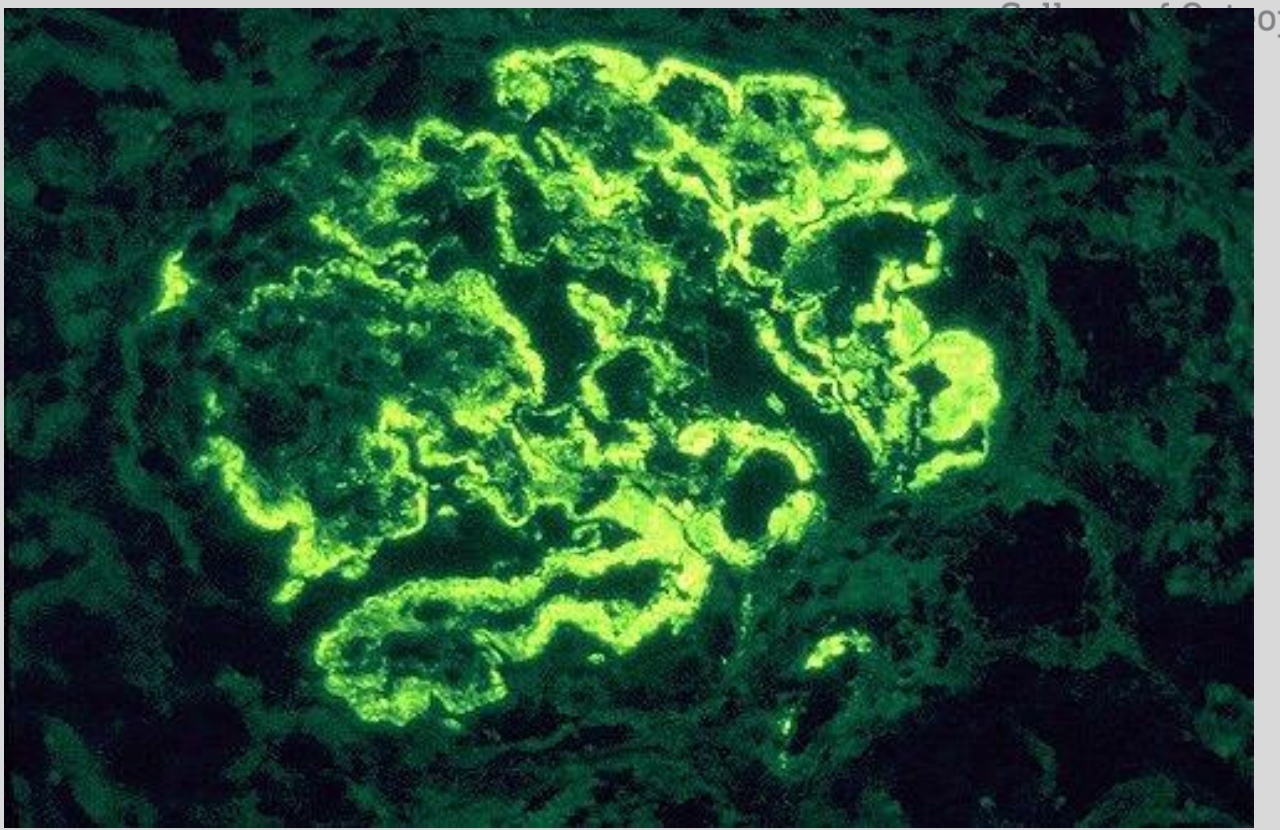


Membranous Nephropathy - prominence of **basement membranes**

MGN/Nephropathy IF

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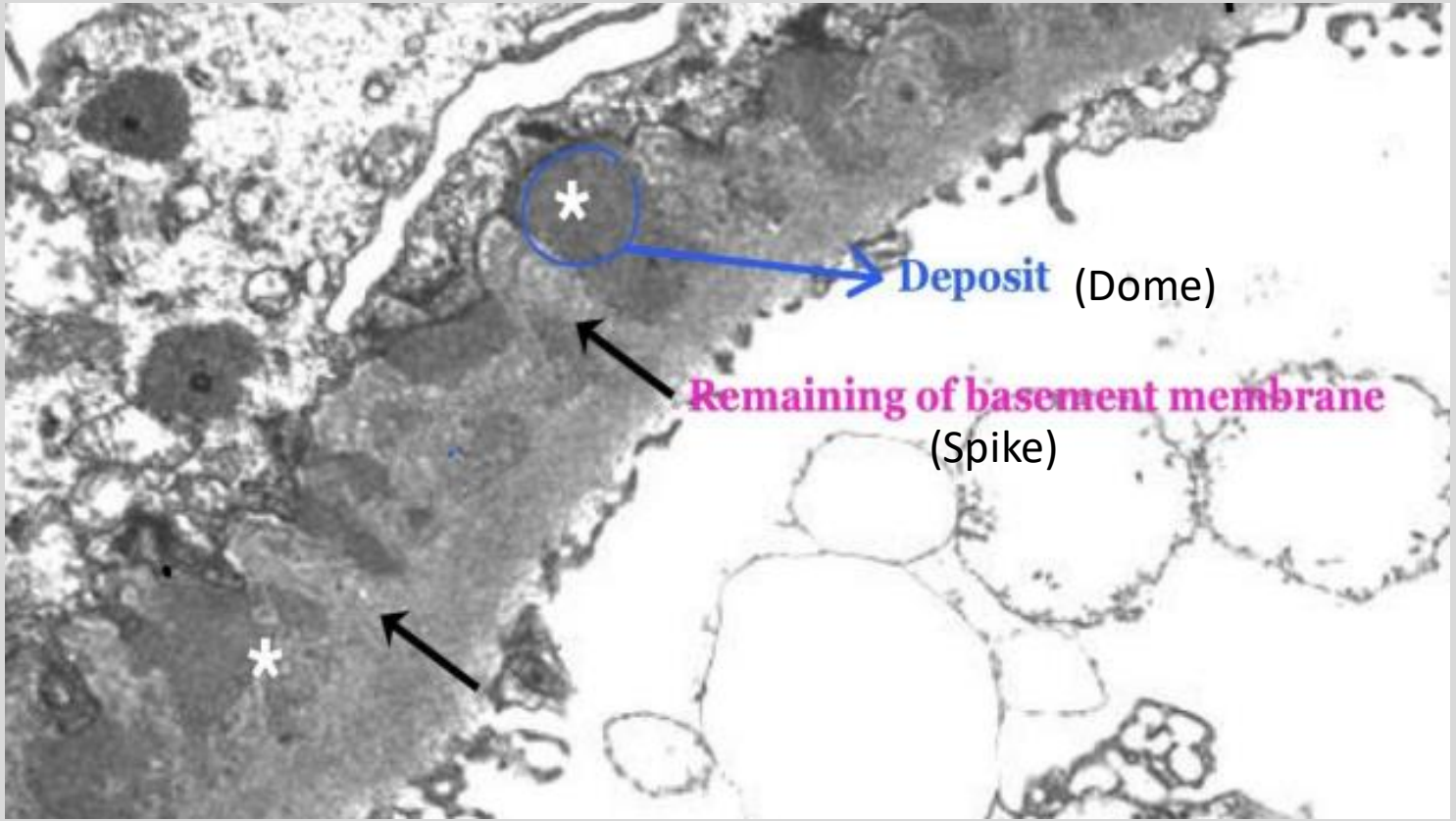
Granular deposits



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MGN/Nephropathy EM

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MGN/Nephropathy clinical presentation

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- Highest reported incidences of deep vein thrombosis (**DVT**), pulmonary embolism (**PE**), renal vein **thrombosis**
  - Proteinuria is **NON-selective**
  - **Hypertension**
  - **Steroid** therapy does **not change survival** rate
  - 1/3 will get better, 1/3 will persist, 1/3 will progress to **ESRD**

- Example presentation
  - **44** year-old **man** with a Hx of **hepatitis B** presents w/ **edema**, and "**foamy** urine"
    - ROS: (+) Weight Gain
    - PE: (+) **Ascites**, (+) Diffuse **Pitting Edema,** (+) **RUQ Pain** (liver congestion),

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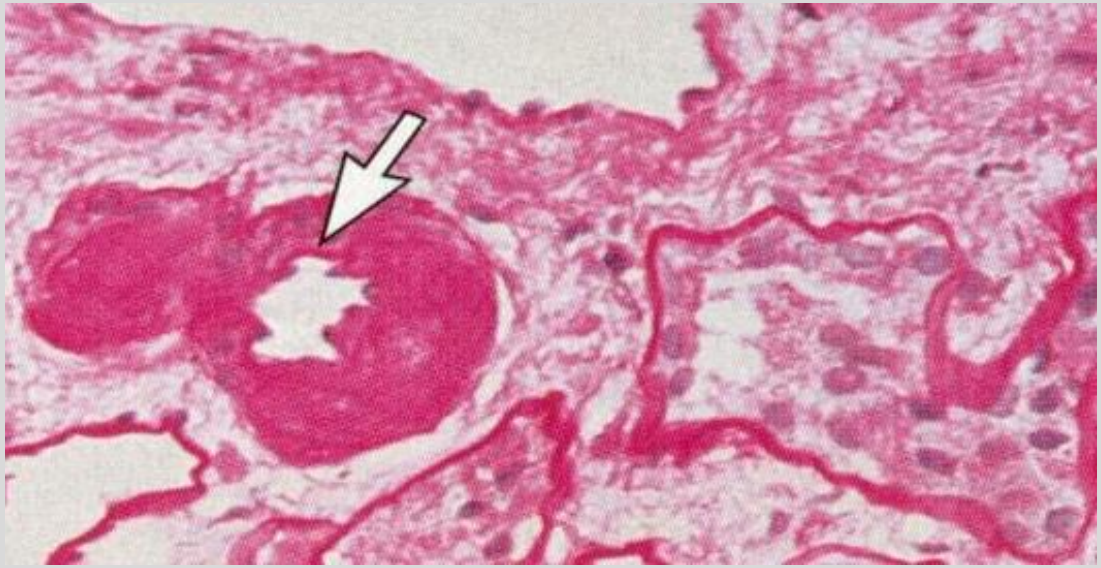
Diabetic Glomerulopathy pathogenesis/pathology

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- Associated with **uncontrolled** Type I (ages 15-25), or Type II (commonly > 30 year of age) **DM**
- Common Sx/Sx
  - Most common cause **chronic renal failure** in US
  - Glomerulopathy & retinopathy, hypertension
- Pathogenesis
  - **Nonenzymatic glycosylation** is caused by high serum glucose & produces arteriolar (afferent & efferent) hyalinosis
  - Alterations in **GBM** is due to hyperfiltration & high pressure in the glomerulus that leads to ↑ Type IV **collagen**→ diabetic **microangiopathy**, glomerular hypertrophy & glomerular **scarring**
- Laboratory findings
  - Initially should detect **micro-albuminuria**
    - If uncontrolled→ macro-albuminuria
  - LM: **Arteriolar hyalinosis** (thickening), **GBM thickening**
    - **Mesangial** matrix nodular masses → trchrome & PAS stain → **Kimmelstiel-Wilson nodules**
  - IF: Nonspecific
  - EM: **Fusion/effacement** of podocytes foot processes

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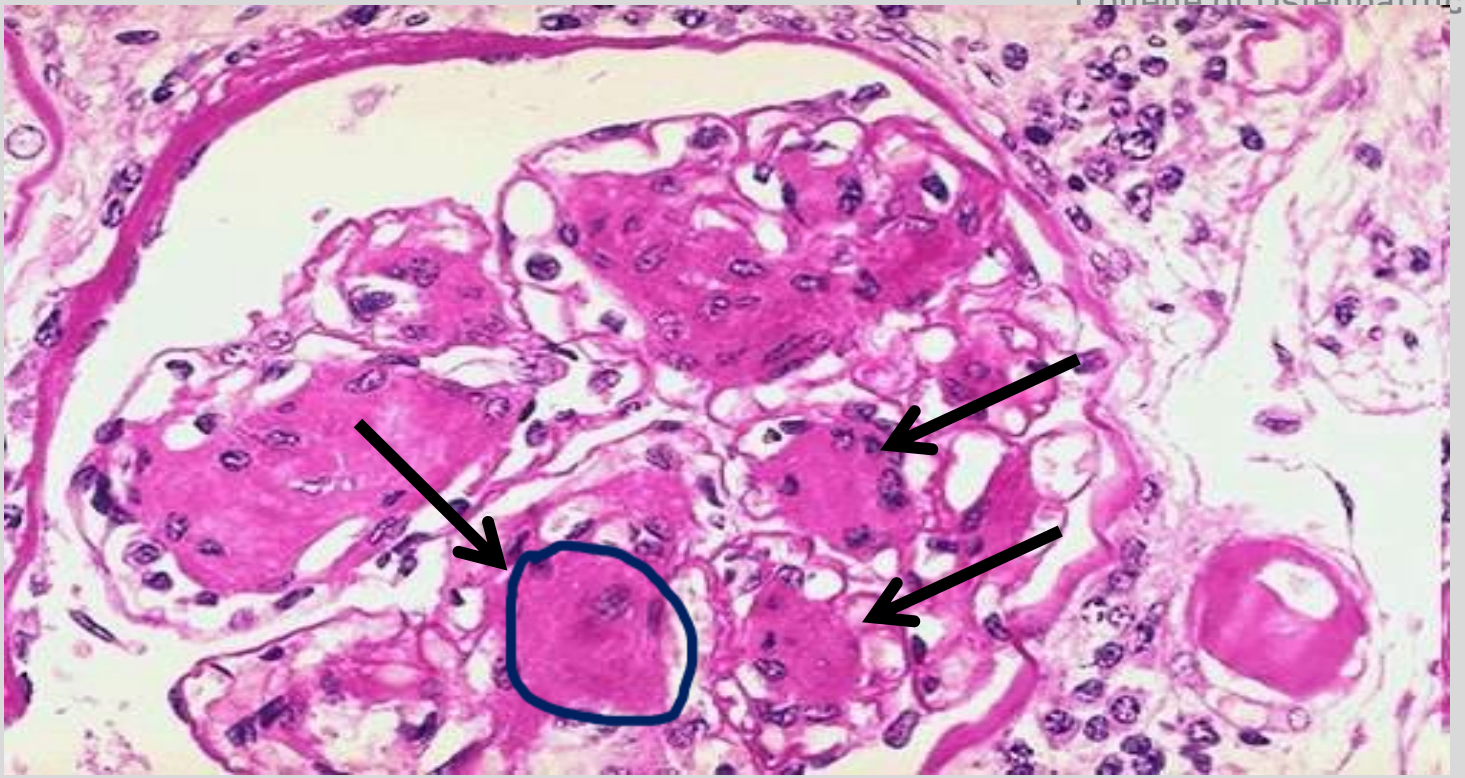
Arteriolar hyalinosis LM



PAS stain

Kimmelstiel- Wilson nodules LM

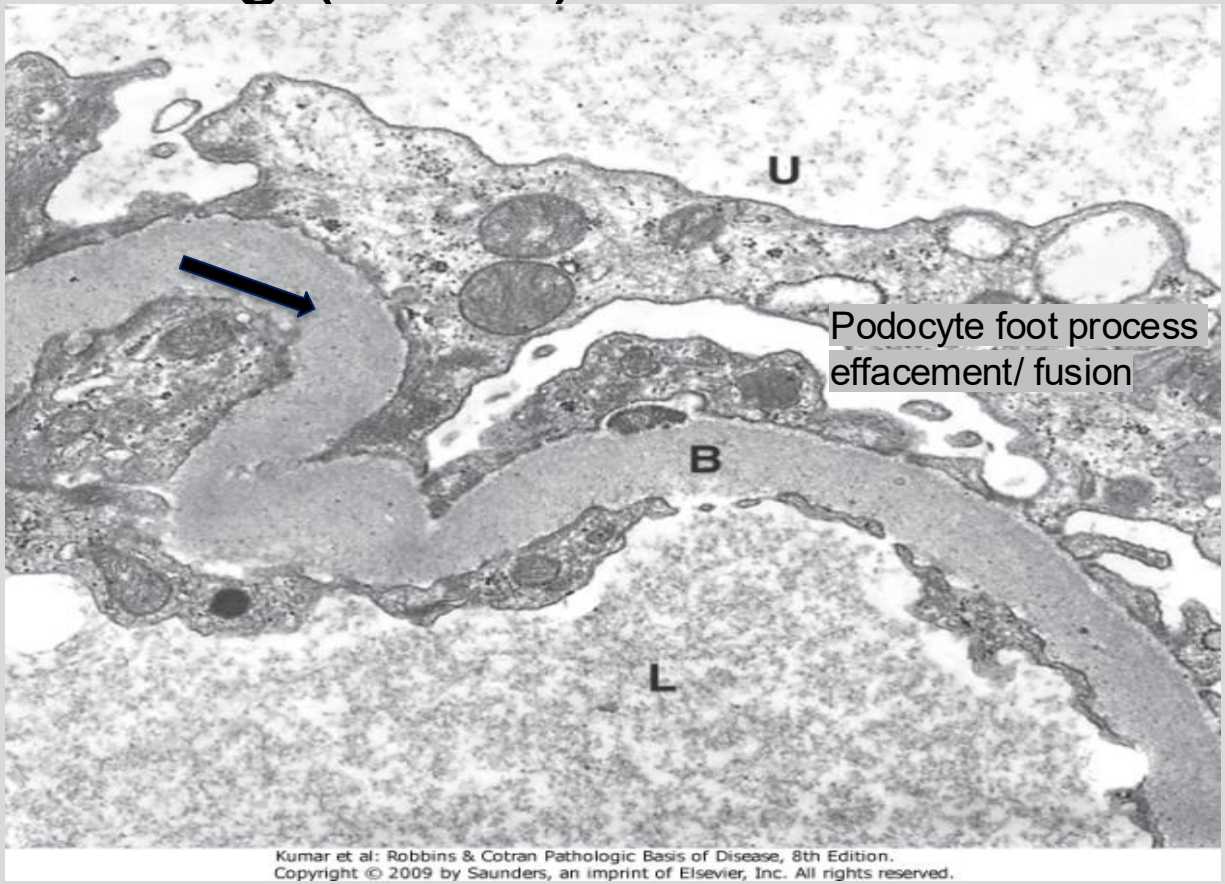
H&E stain



Diabetic Glomerulopathy GBM thickening (arrow)

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Podocyte foot process effacement/ fusion

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Diabetic Glomerulopathy clinical example

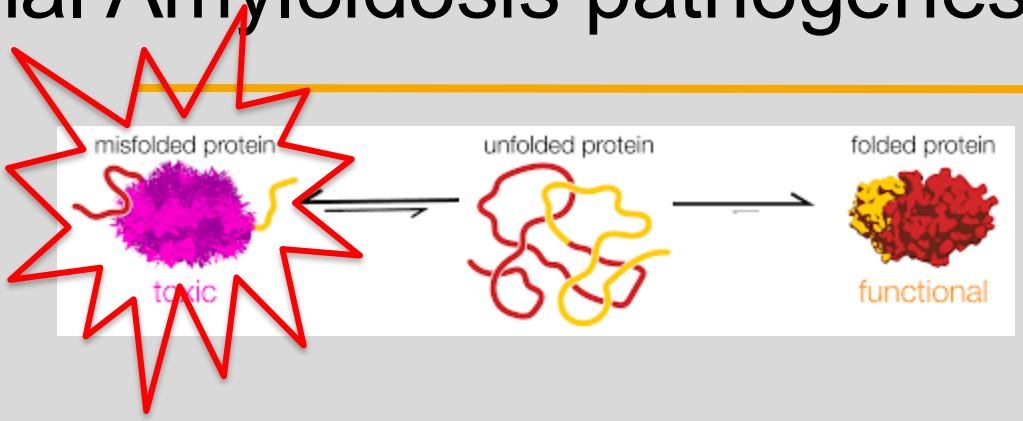
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- **66**-year-old female with a history of **Type II DM** presents for a regular check up, despite not seeing a physician for sometime. Her labs reveal a recent high **rise in lipids**, **albumin** is **low**, and PT/INR are normal
  - UA shows **heavy protein** w/ slight blood
  - ROS: (+) Weight gain
  - On PE: **(+) Anasarca**, (+) Decreased **vision**

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Renal Amyloidosis pathogenesis

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- >20 different forms of amyloidosis, each due to a different **protein misfolding**
- **AL (light chain)**: (**l**ambda>kappa-use special stain, immunoreactivity may be absent)-(think multip**AL** myeloma)
- **AA (inflammatory)**: (**A**myloid of **A**rthritis, chronic inflam**AA**tion)
- Amyloid deposits in & disrupt the **mesangium**-results in nephrotic syndrome

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Renal Amyloidosis histopathology

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- **LM: Congo red** stain (pink- orange) w/ **apple green birefringence** w/ polarized light

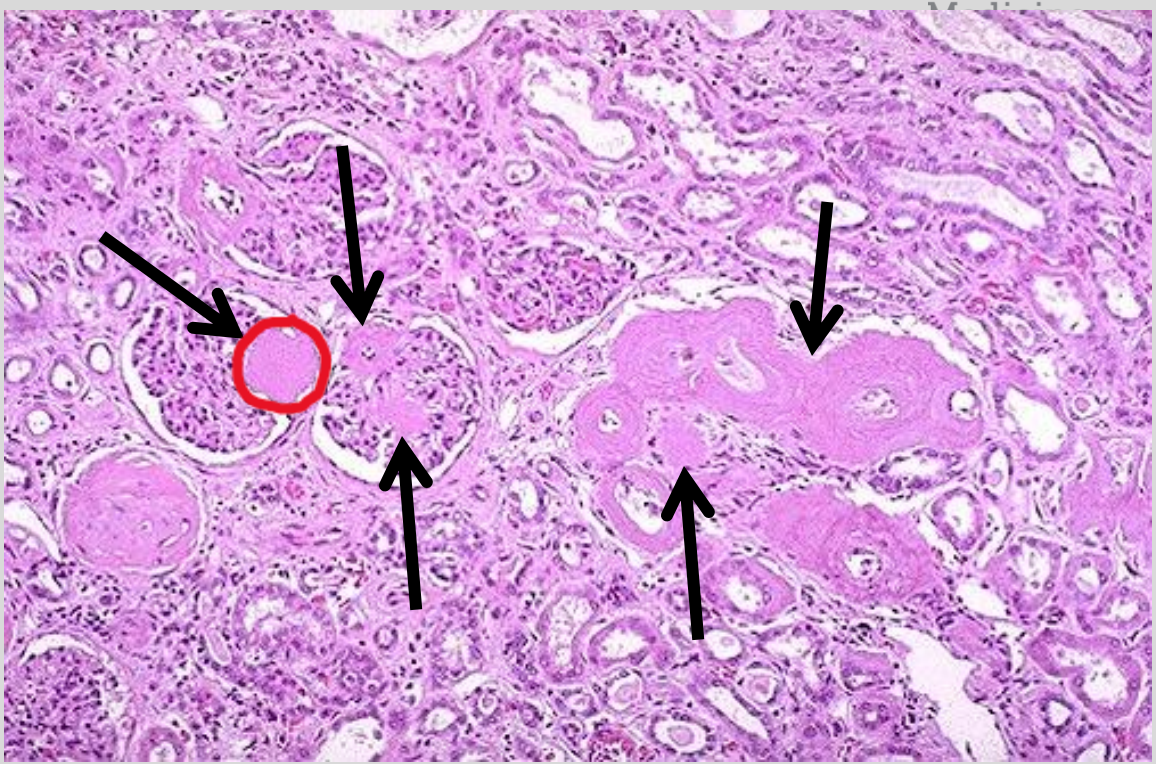
- **EM:** 8-10 nm **fibrils**

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Renal Amyloidosis LM

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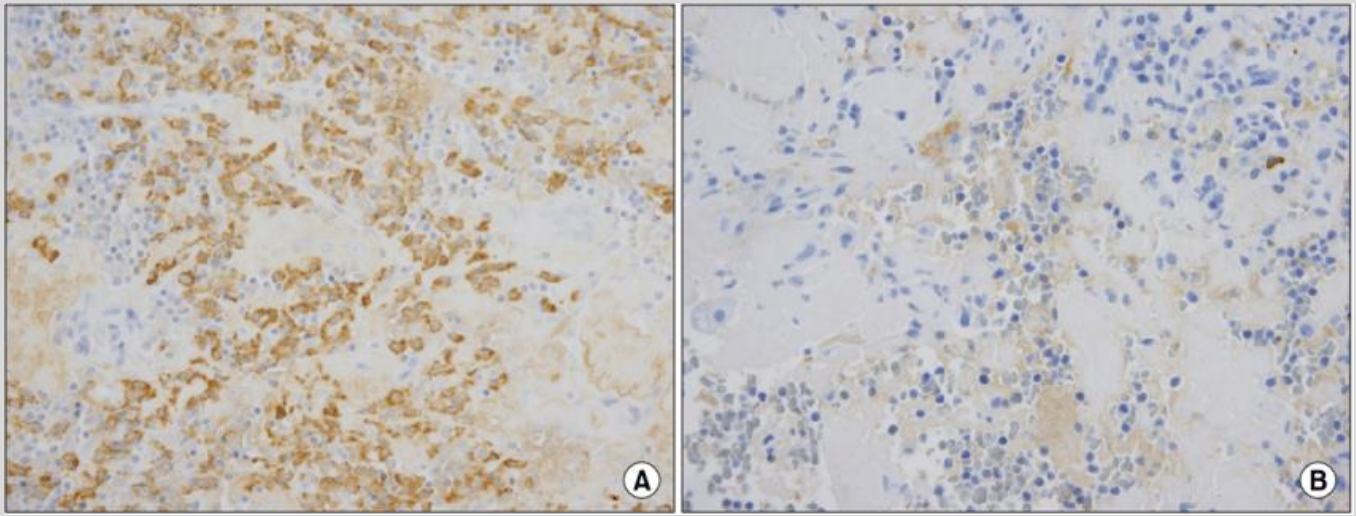
- H&E stained renal tissue
- Amyloid deposits in glomeruli (red circle & arrows on the left) & vessels (arrows on the right)



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Renal Amyloidosis LM

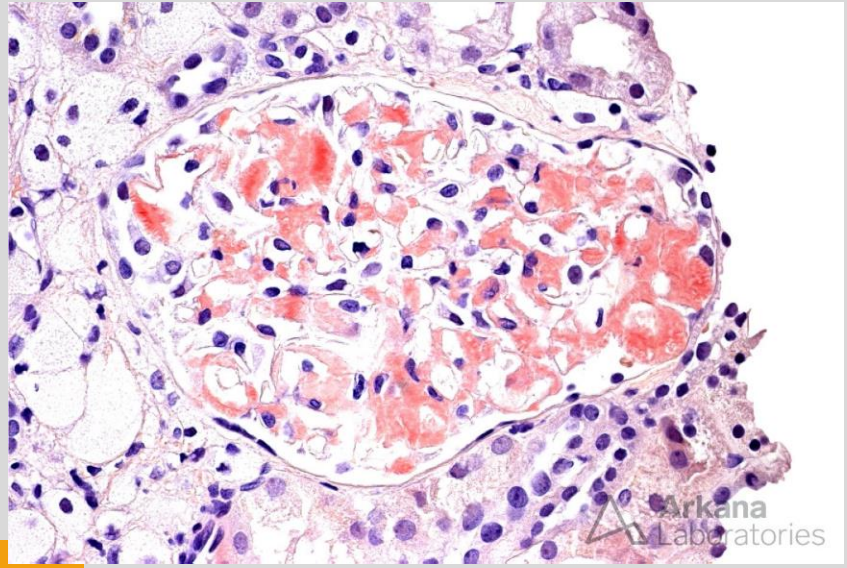
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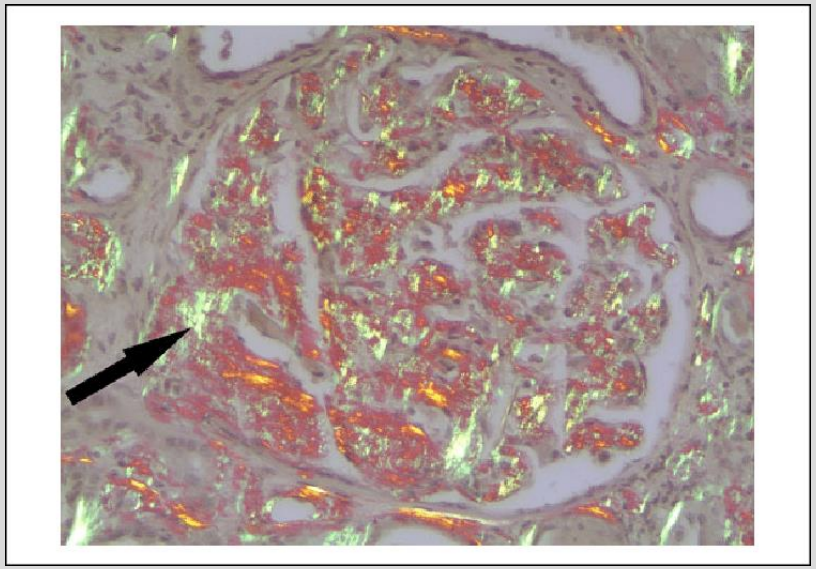


- **Immunohistochemical** stain shows **monoclonality** for lambda light chain (A. positive deposits brown), but negative for kappa light chain (B. nonspecific staining shown)

Renal Amyloidosis LM

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- Special chemical stain: Congo Red w/ salmon colored deposits left
- Polarized LM: apple green birefringence, right

Renal Amyloidosis EM
Fibrils (deposited, cell top left)

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Renal Amyloidosis clinical example presentation

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- **56-year-old** female with history of **monoclonal gammopathy** (in the blood, M protein) & presence of **Bence Jones protein** (urine light chain) in the urine developed **nephrotic syndrome.**

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Membranoproliferative Glomerulonephritis (MPGN)

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- Occurs in **children** & **adults**
- May have **nephritic/nephrotic** presentation
- **Not** be confused w/ **membranous GN**, a condition in which the basement membrane is thickened, but the mesangium is not

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MPGN pathogenesis

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Type I → immune complex (**IC**) **type** of **MPGN**:

- Associated w/ **HCV** (main hepatitis C associated nephropathy)
  - Usually w/ **cryoglobulinemia**
- Inciting antigen **unknown**
- Pathogenesis: ICs activate **complement** pathways

Type II→ aka **Dense Deposit Disease**: Associated w/ **C3 nephritic factors** → autoantibody binds alternative pathway C3 convertase (C3bBb) & prevents degradation of C3 convertase→ overactivation of **compliment**

- Pathogenesis & laboratory findings: Activation of **alternative** complement pathway w/ ↓**serum C3**, ↓**factor B & properdin** (depleted) & **normal C1** & **C4**

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MPGN histopathology

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- Type I & Type II
- LM:
  - **GBM thickening** with ↑ **mesangial** and/or **endocapillary cellularity**
  - "**Tram tracks**" (a train runs on its own tracks separate from other vehicles; a tram shares at least part of its track network with other vehicles such as cars) seen w/ **silver stain** (stains basement membranes well)
    - Also visible with H&E & PAS



A tram on it's tracks

MPGN pathology

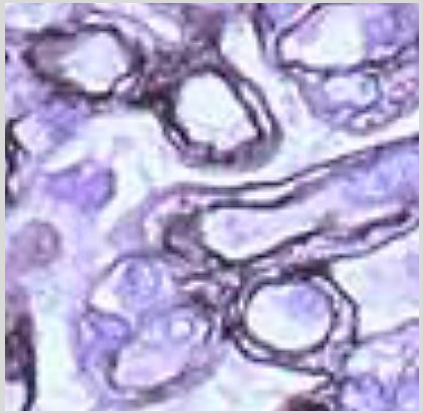
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- Type I
  - IF:
    - Pattern: **Granular**
    - Type of protein deposit:
      - **C3**
      - **IgG**
      - **C1q & C4**
  - EM:
    - **Subendothelial** deposits
    - **Mesangial** deposits
- Type II
  - IF:
    - Pattern: **Granular**
    - Type of protein deposit:
      - **C3 & properdin** are **present**
      - **IgG, C1q & C4** are usually **absent**
  - EM:
    - Diffuse **intra- membranous** déposits:
      - **“Dense deposit disease”**
      - **Ribbon-like** appearance

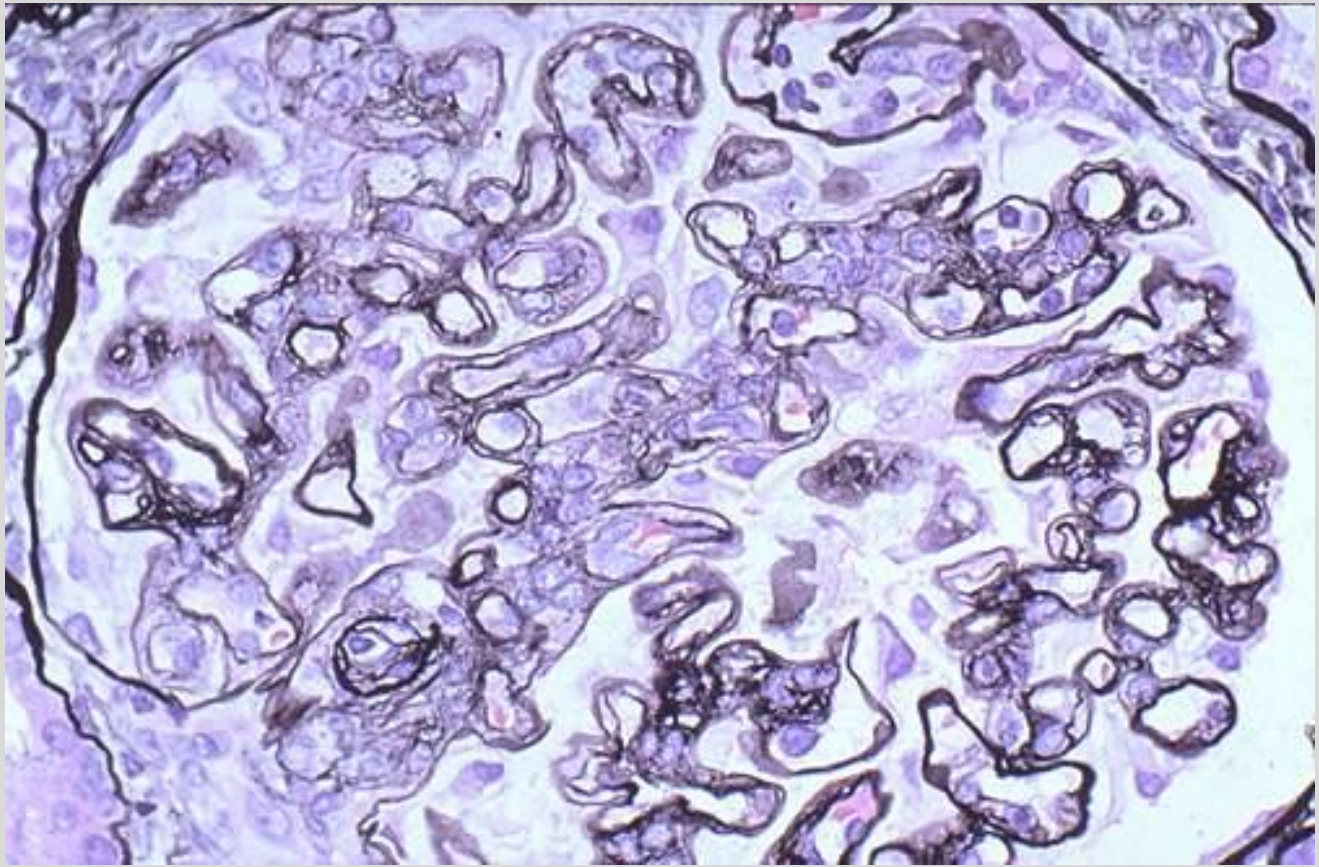
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MPGN LM, silver stain for basement membranes

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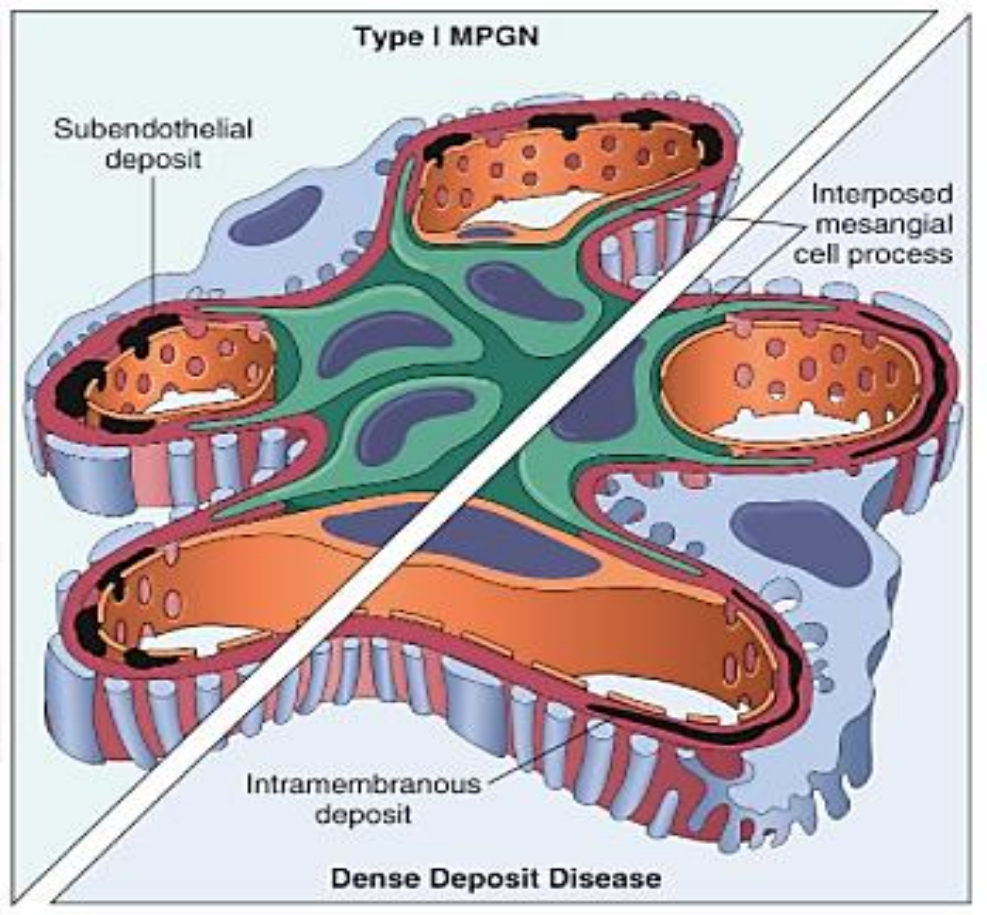






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MPGN



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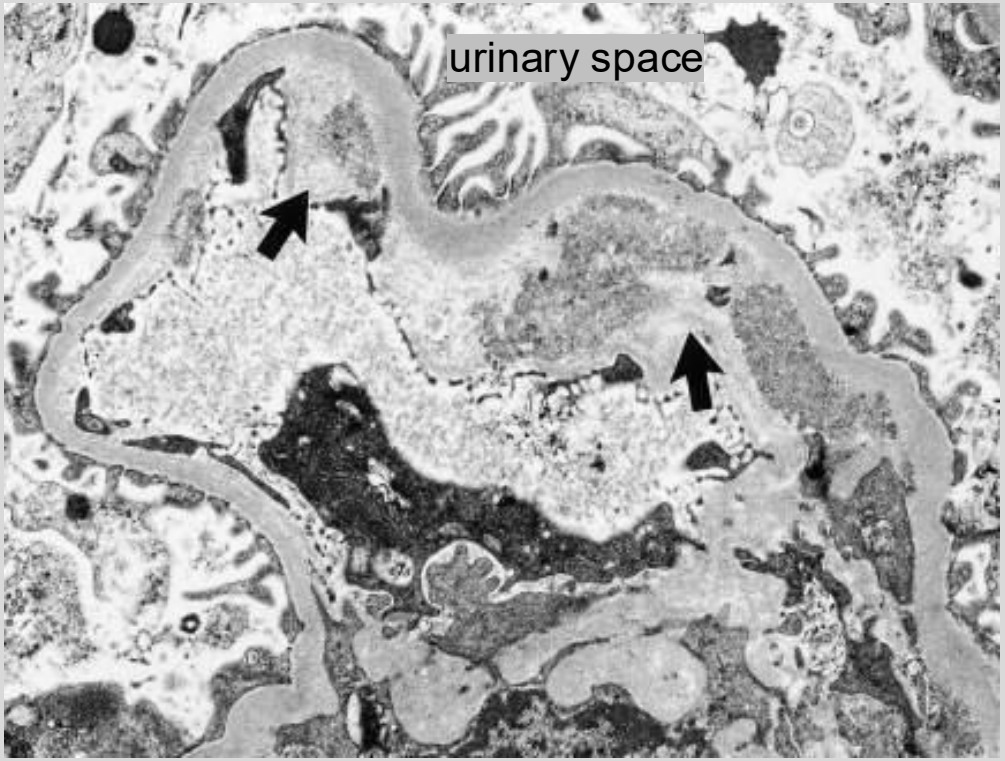
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MPGN Type I EM

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The arrows indicate subendothelial deposits

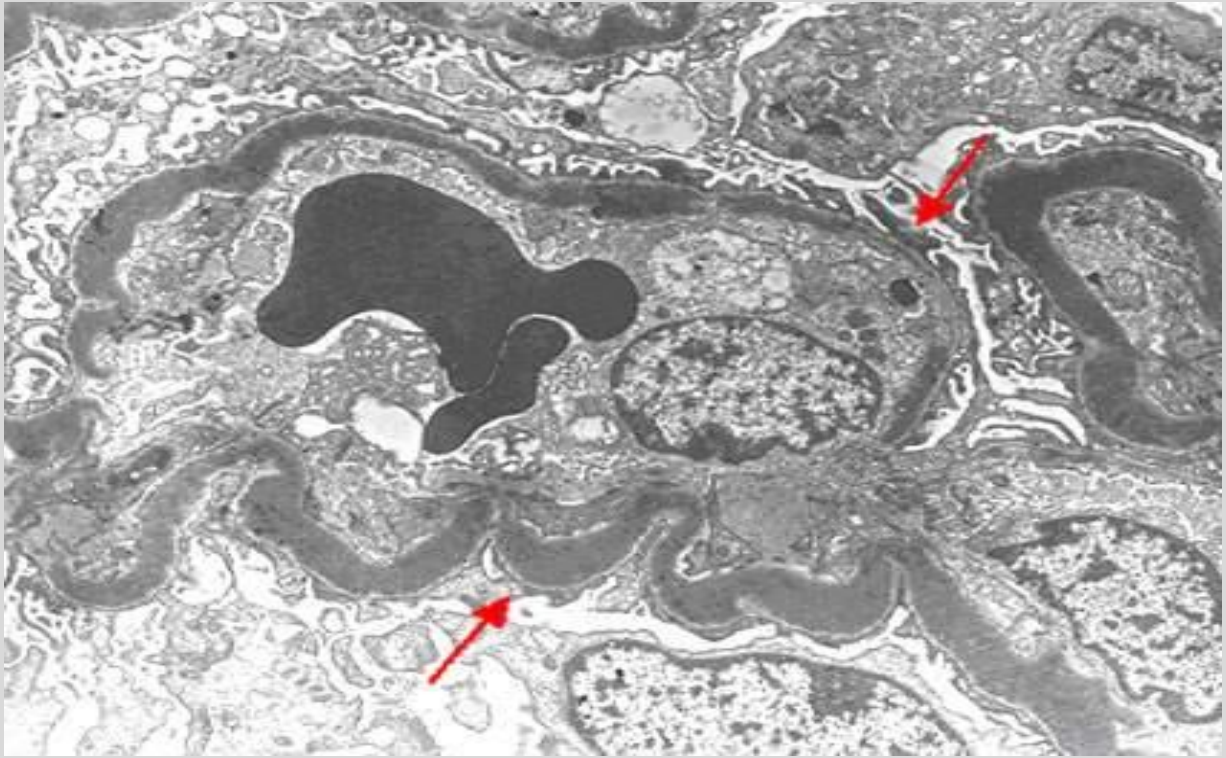


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MPGN Type II – EM

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Ribbon-like appearance of intra-membranous deposits split the basement membranes (tram track appearance)



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MPGN clinical presentation

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- **Variable** presentation w/ nephritic, nephrotic, **nephritic-nephrotic** syndrome
- Most patients have **hypertension**
- Majority **progress** to chronic kidney disease, **CKD**

- Example presentation: **56-year-old** man with a history of **Hepatitis C** presents w/ diffuse **edema** & “dark **foamy** urine”
  - ROS: (+) Weight Gain
  - PE: (+) Ascites, (+) RUQ Pain, (+) Edema
  - UA: (+) **Blood** w/ RBC casts, **3.7g protein**

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Thank you,

Lecture Feedback Form Link:

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